

BERGEN FLESLAND, Norway

WMO: 013110

Lat: 60.293N

Lon: 5.218E

Elev: 52

StdP: 100.70

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-7.2	-5.5	-14.9	1.0	-4.0	-12.2	1.3	-2.9	14.5	4.5	13.3	4.5	1.6	100	0.654

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.8	23.6	16.4	21.4	15.6	19.8	14.8	17.7	21.5	16.7	19.9	15.8	18.6	3.7	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	11.6	18.8	15.4	11.0	17.9	14.5	10.4	17.0	49.9	21.5	46.9	20.0	44.4	18.7	21.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.3	10.6	9.3	-8.8	26.6	2.1	2.3	-10.2	28.3	-11.4	29.7	-12.6	31.0	-14.1	32.7		
(5)			-9.6	19.0	2.2	1.2	-11.2	19.9	-12.5	20.6	-13.7	21.3	-15.2	22.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.6	1.8	1.6	2.9	6.2	9.4	12.2	14.7	14.8	12.2	8.2	4.7	2.5		
	DBStd	5.67	3.32	3.12	2.66	2.93	3.21	2.67	2.79	2.64	2.60	2.97	3.39	3.79		
	HDD10.0	1356	253	234	219	120	49	7	1	0	8	69	161	233		
	HDD18.3	3920	512	468	478	364	278	187	118	116	185	313	410	492		
	CDD10.0	491	0	0	0	6	31	71	147	148	73	14	1	0		
	CDD18.3	14	0	0	0	0	1	2	6	6	1	0	0	0		
	CDH23.3	69	0	0	0	0	3	7	39	21	1	0	0	0		
	CDH26.7	7	0	0	0	0	0	1	6	1	0	0	0	0		
	WSAvg	3.9	4.5	4.2	4.1	3.8	3.8	3.7	3.3	3.4	3.6	4.0	3.9	4.3		
	PrecAvg	2026	175	138	151	107	100	124	137	173	259	240	228	213		
(15) Precipitation	PrecMax	2932	399	336	390	272	204	249	271	344	505	502	384	344		
(16) PrecipMin		1434	44	5	20	11	29	6	26	6	62	50	78	36		
(17) PrecStd		386	94	84	94	52	50	58	66	81	106	104	91	98		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.9	8.8	11.5	18.1	23.0	24.2	27.4	26.0	21.7	17.6	12.7	10.4		
		MCWB	6.1	6.8	7.0	10.5	14.3	15.5	18.0	18.0	15.4	12.8	9.5	8.6		
	2%	DB	7.9	7.6	8.6	15.2	19.8	21.1	23.9	23.0	19.2	14.9	11.1	9.1		
		MCWB	6.3	6.4	5.7	9.1	12.6	14.3	16.9	16.7	13.9	11.2	9.1	7.9		
	5%	DB	7.1	6.9	7.4	12.9	17.2	18.7	21.3	20.8	17.6	13.3	10.2	8.3		
		MCWB	5.9	5.8	5.5	8.0	11.3	13.2	16.3	16.0	13.6	10.8	8.4	7.2		
	10%	DB	6.2	6.1	6.6	10.8	15.0	16.8	19.3	19.0	16.1	12.2	9.2	7.6		
		MCWB	5.1	5.1	5.0	7.0	10.2	12.5	15.3	15.3	13.0	10.2	7.7	6.4		
	0.4%	WB	7.7	7.6	7.7	11.7	15.4	16.5	19.1	19.0	16.7	13.7	10.9	9.4		
		MCDB	8.0	7.9	8.9	16.5	21.8	22.4	24.7	23.1	19.3	15.3	11.6	9.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	6.9	6.7	6.8	9.8	13.4	15.1	17.8	17.7	15.4	12.6	10.0	8.4		
		MCDB	7.3	7.2	7.7	13.2	18.0	19.4	22.1	21.1	17.7	13.9	10.8	8.8		
	5%	WB	6.2	6.1	6.2	8.7	12.0	14.0	16.8	16.8	14.3	11.7	9.0	7.6		
		MCDB	6.7	6.6	7.1	11.6	16.3	17.4	20.1	19.7	16.5	13.0	9.8	8.1		
	10%	WB	5.3	5.2	5.4	7.8	10.8	13.1	15.9	15.9	13.5	10.7	8.0	6.5		
		MCDB	6.1	5.9	6.3	9.9	14.2	15.8	18.7	18.3	15.5	12.0	8.9	7.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.0	4.4	5.0	6.2	6.6	6.1	5.9	5.8	5.4	5.0	4.3	4.0		
		MCDBR	4.2	4.3	5.7	9.5	10.2	9.9	9.5	8.5	7.5	5.9	4.5	4.2		
		MCWBR	3.9	3.7	3.9	4.9	5.0	4.7	4.2	4.0	4.0	3.7	3.9	4.0		
		MCDBR	4.1	3.9	4.7	7.5	9.3	8.2	8.1	7.3	5.9	4.9	4.3	4.1		
	5% WB	MCWBR	4.0	3.8	3.9	4.3	4.9	4.5	4.1	3.9	3.9	3.9	4.4	4.1		
(40) Clear-Sky Solar Irradiance	taub		0.289	0.319	0.341	0.371	0.360	0.355	0.370	0.367	0.351	0.329	0.295	0.395		
	taud		2.148	2.275	2.332	2.321	2.429	2.475	2.456	2.462	2.494	2.434	2.232	2.609		
	Ebn at Noon		456	671	783	825	867	876	851	825	776	664	452	322		
	Edh at Noon		43	67	87	105	101	99	99	91	75	58	39	30		
(44) All-Sky Solar Radiation	RadAvg		0.23	0.75	1.83	3.45	4.95	5.27	4.67	3.58	2.20	1.07	0.34	0.14		
	RadStd		0.04	0.13	0.28	0.34	0.48	0.63	0.44	0.35	0.27	0.16	0.05	0.02		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	-92	N/A	N/A	N/A
(47) Regional (39 neighbors)		N/A	+0.68	N/A	N/A	N/A	N/A	-84	-123	N/A	N/A

Nomenclature: See separate page